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Article *in* IET Power Electronics · February 2014

DOI: 10.1049/iet-pel.2013.0071

www.ietdl.org

Published in IET Power Electronics
Received on 21st January 2013
Revised on 12th April 2013
Accepted on 11th May 2013
doi: 10.1049/iet-pel.2013.0071

ISSN 1755-4535

Reduced DC voltage source flying capacitor multicell multilevel inverter: analysis and implementation

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Abstract: Voltage source multicell multilevel converters (VSMMCs) have been extensively exploited in medium-voltage high-power fed industrial applications. One of the typical breeds of the VSMMCs is flying-capacitor multicell converter (FCMC). This study presents an improved configuration for FCMCs, accompanied by its analysis, modelling and implementation. The main advantage of the proposed converter, in comparison with the conventional one, is that the number and voltage rating of the required dc voltage sources are halved in the improved topology which results in reducing the cost, size, installation area and weight of the FCMCs effectively. This amelioration is attained by adding four power switches to the conventional topology of an FCMC without making any amendments to the number and voltage rating of high-frequency power-switches and flying capacitors (FCs). The proposed switching and control methodology for the adopted FCMC is based on the phase-shifted carrier pulse width modulation technique with applying some amendments; therefore the natural balancing phenomenon of the FC voltages, one of the critical advantages of the FCMCs, is preserved in the proposed VSMMC. Experimental measurements taken from the 3-cell-four-level and 4-cell-five-level laboratory prototype systems of the proposed converter are presented in order to validate and corroborate the effectiveness and advantages of the proposed topology, its switching and suggested control strategy. Furthermore, experimental studies are accomplished on an 8-cell-nine-level prototype system of the proposed FCMC to peruse its feasibility and viability for higher number of cells and voltage levels. In addition, comparison of the proposed converter with the other types of multilevel converters is carried out to underline the advantages of the proposed structure profoundly.

1 Introduction

On account of the advent of voltage/current source multilevel converter topologies facilitating the power conversion in small steps of voltage/current, subsequent to a series/parallel assembly of medium/low voltage/current rated insulated gate bipolar transistor (IGBT) power switch (currently 8 kV and 6 kA)-based switching cells in order to produce the ultimate voltage/current stepped-waveform, it has been conceivable to realise the high-powered industrial applications such as high power ac motor drives, active power filters, reactive power compensation, compressors, extruders, pumps, fans, grinding mills, rolling mills, conveyors, crushers, blast furnace blowers, gas turbine starters, mixers, mine hoists, flexible alternating current transmission system devices, high voltage direct current applications, dispersed generation and renewable energy integrations to the electric power grid, hydro pumped storage, wind energy conversion and railway traction [1–3].

In comparison with the classic two-level power converters, the most noteworthy advantages of multilevel converters can be indicated as follows: a significant enhancement to the synthesised voltage/current waveforms from the harmonic content point of view, reduced voltage/current rating of the embedded semi-conductor power switches, increased reliance on the converter operation owing to the features such as fault tolerance, decreased switching losses, enhanced efficiency, reduced amount of output d*v*/d*t* or d*i*/d*t* stresses, lowered electromagnetic interference and output filter inductance etc. [4–8].

The most common multilevel converter topologies are the neutral point clamped (NPC) converters, cascaded H-bridge (CHB) converters, flying capacitor (FC) converters and their hybrid topologies [9–12].

The first NPC pulse-width modulation (PWM) converter, also named as the diode-clamped converter, was presented by Nabae *et al.* in 1981. The NPC converters use a serial connection of capacitors at the common dc link with an intention to divide its voltage into different levels. Employing an appropriate switching pattern in a matrix of semi-conductor switches and clamping diodes maps these dc-link voltage levels at the converter output [1, 2].

Years later, the NPC converters matured into a generalised topology referred as multi point clamped (MPC) converters to fulfil the requirements of industry deservedly and to power a wide range of applications. Despite of the well understood technical merits of the MPC converters in the academic arena, they have not reached the medium-voltage market and high-power industrial applications yet owing to sophistication in negating the dc-capacitor voltage drift phenomenon [13].

It is worth mentioning that the 3-level NPC converters are well established in medium-voltage drives and power system applications [14]. Recently, a diode-clamped based five-level 6.6 kV transformerless medium-voltage motor drive for application such as fans, blowers and pumps without regenerative braking has been presented [15]. Hasegawa and Akagi [16] have proposed a single compact coupled inductor, suitable for the five-level diode-clamped converters, as a new dc-voltage-balancing circuit. Shukla *et al.* [17] has implemented a chopper circuit based on FC converter to balance the voltage of dc-link capacitors in diode-clamped converters to improve the reliability of the circuit.

The CHB converters, also referred as cascaded multicell (CM) converters, use a series connection of three-level H-bridge modules to obtain a staircase-like output voltage. Despite of requiring multiple isolated dc voltage sources which causes bulky and costly input transformers with several phase shifted secondary windings, CHB converters have been successfully commercialised for medium-voltage high-power applications owing to their modular configuration that eases series expansion [8, 10, 18–21].

Ebrahimi *et al.* [8] has proposed a new CM converter utilising single-phase sub-multicell converter units which are based on H-bridge cells. The significant feature of the proposed configuration is reduction in the number of required high-frequency power switches that result in decreasing the size and cost of the circuit [8]. Khoshkbar Sadigh *et al.* [4] has proposed a new asymmetrical CM (ACM) converter based on optimised symmetrical sub-multilevel modules resulting a significant reduction in the number of required high-frequency power switches and associated driver circuits that implies cost, weight and installation area enhancements to CHB converters.

Other significant topologies of multilevel converters are FC multicell converters (FCMCs), and their sub-topologies stacked multicell (SM) converters (SMCs) [11, 12, 22]. These converters are based on the interconnection of cell units which consists of a FC and a pair of semi-conductor switches with a complementary state [23]. Redundant switching states in the FCMCs and SMCs permit the stabilisation of voltage across the FCs at their requisite values [24]. The commutation between the adjacent cells with their associated FCs charged to the required voltage values generates different levels of chopped input voltage at the output side of the aforementioned converters [22].

Khoshkbar Sadigh *et al.* [25] has proposed a new configuration for FCMCs, referred as double FCMCs (DFCMCs), which do possess an advantage of the reduced number of high-frequency power switches and FCs by a value of 50% as a consequence of entailing two additional low-frequency switches. Recently, Dargahi *et al.* [26] has proposed an improved topology for DFCMCs, referred as improved DFCMCs (I-DFCMCs), in order to make the number of required FCs and their voltage ratings to descend significantly implying cost, weight and installation area enhancements to FC converters.

Recently, several studies have proposed hybrid multicell converters based on the combinations and variations of the well-known topologies such as five-level FCMC-based active neutral-point-clamped (ANPC) converter by Pulikanti *et al.* [27], seven-level converter based on the upgrade of the five-level ANPC converter by Saeedifard *et al.* [28], five-level ANPC converter in which common dc link for three phases is composed of six switches in a crossed configuration and one capacitor by Chaudhuri *et al.* [29].

In other side from control methodology point of view, numerous papers have investigated different types of modulation methods [30–34]. McGrath and Holmes [31] has studied and analysed the operation of FCMC when it is controlled by a phase-disposition PWM (PD-PWM). Shukla *et al.* [32] has proposed a new strategy to achieve natural voltage balancing for FCMC when it is controlled by PD-PWM. In addition, the same authors have studied and reviewed the various hysteresis modulation approaches for multilevel converters considering their pros and cons [33]. She *et al.*, [34] in order to balance the voltage of the cells in a cascaded seven-level rectifier stage of solid state transformer, has proposed a three dimension space modulation technique with voltage balancing capability.

An inherent mechanism pertinent to the balance of voltage across the FCs in the FCMCs, well-recognised as natural balancing property, is analysed by mathematical modelling in literatures [35–39]. This phenomenon results in safe operation of the converter and also, equal distribution of voltage stress on semi-conductor switches [40]. The aforesaid flying-capacitor-based multicell converters have attained a significant interest over recent years owing to their noteworthy advantages such as no need for isolated dc links, transformerless operation capability, no need for clamping diodes, availability of redundant states to balance the voltage across the FCs, inherent natural balancing property of the FC voltages, and equal distribution of switching stress on semi-conductor switches [1–3, 9, 11, 12, 17, 22–27, 30–32, 35–45].

Addressing the above mentioned features, any effort to improve the topology or control methodology of the flying-capacitor-based converters such as FCMC, SMC, DFCMC, I-DFCMC and ANPCFC, make them more practical and efficient in medium-voltage high-power applications. Hence this paper proposes an improved topology for the FCMCs. The main advantage of the proposed topology, as compared with the conventional FCMC, is that the latter requires two dc voltage sources whereas the new structure requires only one of those two dc voltage sources in order to produce an identical voltage at the converters' output stage (i.e. the same number of voltage levels and rms value at the converter output).

In other words, the number and voltage rating of the required dc voltage sources are halved in the improved topology which results in reducing the cost, size, installation area and weight of the FCMCs effectively. This amelioration is attained by adding four power switches to the conventional topology of an FCMC without making any amendments to the number and voltage rating of high-frequency power-switches and FCs. The proposed switching and control methodology for the adopted FCMC is based on the phase-shifted carrier PWM (PSC-PWM) technique with applying little amendments; therefore the natural balancing phenomenon of the FC voltages, one of the critical advantages of the FCMCs, is preserved in the proposed voltage source multicell multilevel converters (VSMMC).

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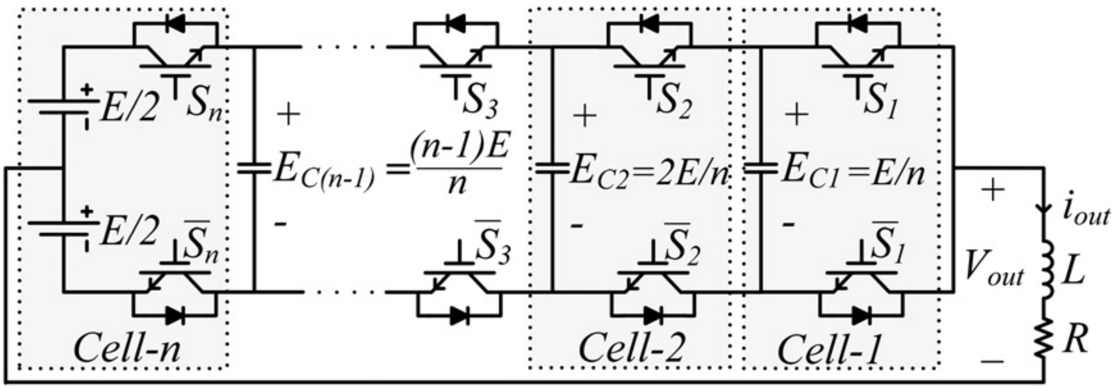


Fig. 1 *n-cell-n + 1-level FCMC producing the output voltage with peak-to-peak value of E*

The remainder of this paper is organised as follows. Section 2 provides a brief summary of the conventional FCMC topology and its properties. In Section 3, the improved configuration of the FCMC and its suggested control strategy which is based on the PSC-PWM technique are discussed. Experimental measurements taken from two prototype systems of the proposed FCMC, that is, 3-cell-four-level and 4-cell-five-level converters are presented in Section 4 to corroborate the feasibility of the proposed topology. Furthermore, experimental studies are accomplished on an 8-cell-nine-level prototype system of the proposed converter to investigate its feasibility for higher number of cells and voltage levels in Section 4. In addition, comparison of the other types of multilevel converters with the proposed FCMC to produce the identical output voltage (i.e. the same number of levels and peak to peak voltage), by considering the number of required components, is illustrated in Sections 5 and 6.

2 Preliminaries on conventional configuration of an FCMC

A typical configuration of an FCMC, as shown in Fig. 1, is realised by a tandem connection of n controllable modular structures generically called cells which are connected in series to form required converter leg and can produce $n+1$ levels of voltage with peak to peak value of E at the output. Each cell in FCMC is made-up by one FC and a pair of power switches possessing complementary states.

It is worth mentioning that because of the identical output current the capacitance of FCs are the same in order to obtain the same voltage ripple. However, FC dc voltage ratings are different and equal to E/n, $2E/n$, …, $(n-1)E/n$ [31–35]. As a result, each power switch sustains just a fraction of dc-link voltage, that is, E/n, and the energy stored in the ith capacitor is [16]

$$U_i = \frac{1}{2}C\left(\frac{iE}{n}\right)^2 \qquad (1)$$

where C and E are the capacitance of FCs and total voltage rating of dc sources, respectively.

As an example, Fig. 2 shows the control strategy, switches states, and the output voltage of a 4-cell-five-level FCMC controlled by the PSC-PWM and operated with a modulation index equal to 0.8 ($M=0.8$). The power switch is ON when its state is 1 and is OFF when its state is 0. State of power switches of the 4-cell-five-level FCMC are also illustrated in Table 1.

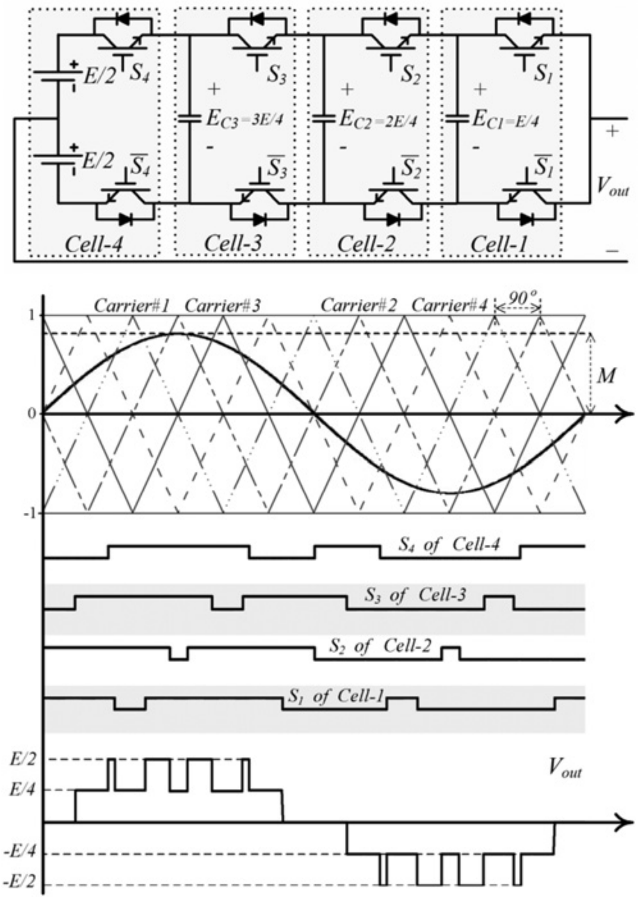


Fig. 2 *4-cell-five-level FCMC, its control strategy based on PSC-PWM, states of power switches and output voltage*

Table 1 States of power switches in a 4-cell-five-level FCMC

Output voltage level	State of switches (S_4, S_3, S_2, S_1)	Number of States
$+0.5E$	(1, 1, 1, 1)	1
$+0.25E$	(1, 1, 1, 0) (1, 1, 0, 1) (1, 0, 1, 1) (0, 1, 1, 1)	4
0	(1, 1, 0, 0) (1, 0, 0, 1) (0, 0, 1, 1) (1, 0, 1, 0) (0, 0, 1, 1) (0, 1, 0, 1)	6
$-0.25E$	(1, 0, 0, 0) (0, 1, 0, 0) (0, 0, 1, 0) (0, 0, 0, 1)	4
$-0.5E$	(0, 0, 0, 0)	1

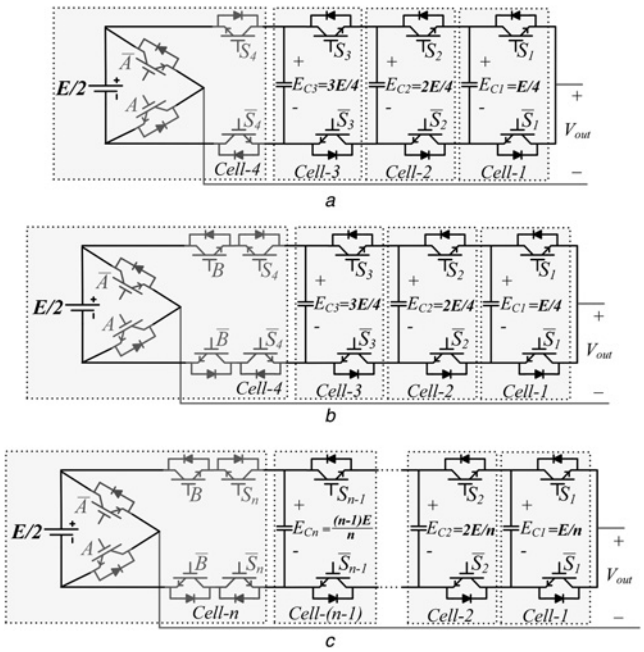


Fig. 3 *Proposed FCMC based on only one dc voltage source with maximum peak-to-peak voltage of E*

a First design of the 4-cell-five-level converter
b Second design of the 4-cell-five-level converter
c Generalised configuration of the *n*-cell-*n* + 1-level converter

To satisfy the natural voltage balancing of the FCs, a regular phase shift must be applied to the control signals. This fact is illustrated in Fig. 2 for 4-cell-five-level FCMC controlled by the PSC-PWM switching technique.

As shown in Fig. 2, the phase shift between the carriers of adjacent cells is [15, 16]

$$\delta = \frac{2\pi}{n} \tag{2}$$

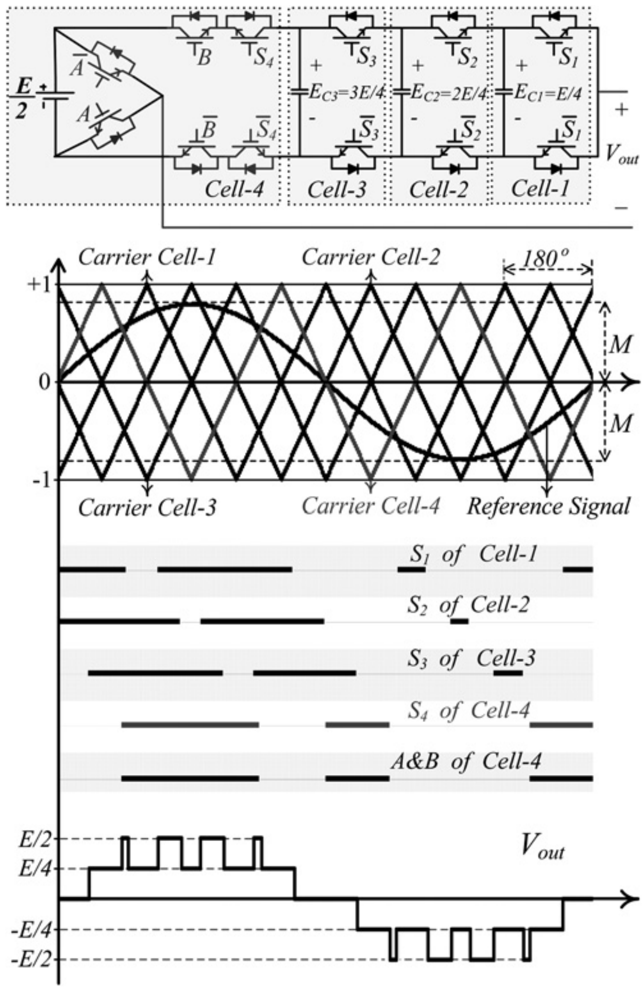


Fig. 4 *4-cell-five-level proposed FCMC, its switching pattern based on the suggested PSC-PWM, states of the power switches and output switched voltage*

Table 2 States of power switches and charging/discharging process of FCs in the 4-cell-five-level proposed FCMC

Output voltage level	State of switches (A, B, S_4, S_3, S_3, S_1)	ΔE_{C3}	ΔE_{C2}	ΔE_{C1}	i_{Load}	Number of states
$+E/2$	(1, 1, 1, 1, 1, 1)	0	0	0	whatever	1
$+E/4$	(1, 1, 1, 1, 1, 0)	0	0	+	$i_{Load} > 0$	4
		0	0	−	$i_{Load} < 0$	
	(1, 1, 1, 1, 0, 1)	0	+	−	$i_{Load} > 0$	
		0	−	+	$i_{Load} < 0$	
	(1, 1, 1, 0, 1, 1)	+	−	0	$i_{Load} > 0$	
		−	+	0	$i_{Load} < 0$	
	(0, 0, 0, 1, 1, 1)	−	0	0	$i_{Load} > 0$	
		+	0	0	$i_{Load} < 0$	
0	(1, 1, 1, 1, 0, 0)	0	+	0	$i_{Load} > 0$	4
		0	−	0	$i_{Load} < 0$	
	(1, 1, 1, 0, 0, 1)	+	0	−	$i_{Load} > 0$	
		−	0	+	$i_{Load} < 0$	
	(0, 0, 0, 0, 1, 1)	0	−	0	$i_{Load} > 0$	
		0	+	0	$i_{Load} < 0$	
	(0, 0, 0, 1, 1, 0)	−	0	+	$i_{Load} > 0$	
		+	0	−	$i_{Load} < 0$	
$-E/4$	(0, 0, 0,0, 0, 1)	0	0	−	$i_{Load} > 0$	4
		0	0	+	$i_{Load} < 0$	
	(0, 0, 0, 0, 1, 0)	0	−	+	$i_{Load} > 0$	
		0	+	−	$i_{Load} < 0$	
	(0, 0, 0, 1, 0, 0)	−	+	0	$i_{Load} > 0$	
		+	−	0	$i_{Load} < 0$	
	(1, 1, 1, 0, 0, 0)	+	0	0	$i_{Load} > 0$	
		−	0	0	$i_{Load} < 0$	
$-E/2$	(0, 0, 0, 0, 0, 0)	0	0	0	whatever	1

doi: 10.1049/iet-pel.2013.0071

Table 3 Power circuit parameters used in experimental observation of the proposed FCMC inverter

System parameters	Values
main dc voltage source rating (*E/2*)	300 V
internal FCs (*C*)	680 uF
PSC-PWM carrier frequency (f_{SW})	5 kHz
modulation index (M)	0.8
fundamental output voltage frequency	50 Hz
resistive-inductive load (*R–L*)	26 Ω–30 mH
balance booster circuit (R_b–L_b–C_b)	90 Ω–6.97 uH – 145 uF

where n is the number of cells and equals to 4 in the 4-cell-five-level FCMC.

The voltage balancing of FCs which ensures the safe operation of the flying-capacitor-based multicell converters is a concern of utmost importance in these topologies. It is well demonstrated by mathematical modelling of FCMCs that the capacitors' voltage balancing (natural balancing) occurs if the PSCPWM or PDPWM techniques are applied to the converter's control unit; in addition, demonstrating that the PDPWM switching technique is far superior to the PSCPWM modulator in balancing the FC voltages.

Since the natural balancing of the flying-capacitor-based converters relies on the current ripple of the output current, the balancing mechanism might be too weak for practical use in some applications where the load impedance at the switching frequency and above is high and the total current ripple is consequently low. For such applications, notably drive applications, a balance booster, a notch filter with low impedance at the switching frequency is often connected in parallel with the load to increase balancing. The output RLC balance booster circuit should be tuned to resonate at the switching frequency as follows

$$\sqrt{L_b \times C_b} = \frac{1}{2 \times \pi \times f_{SW}} \tag{3}$$

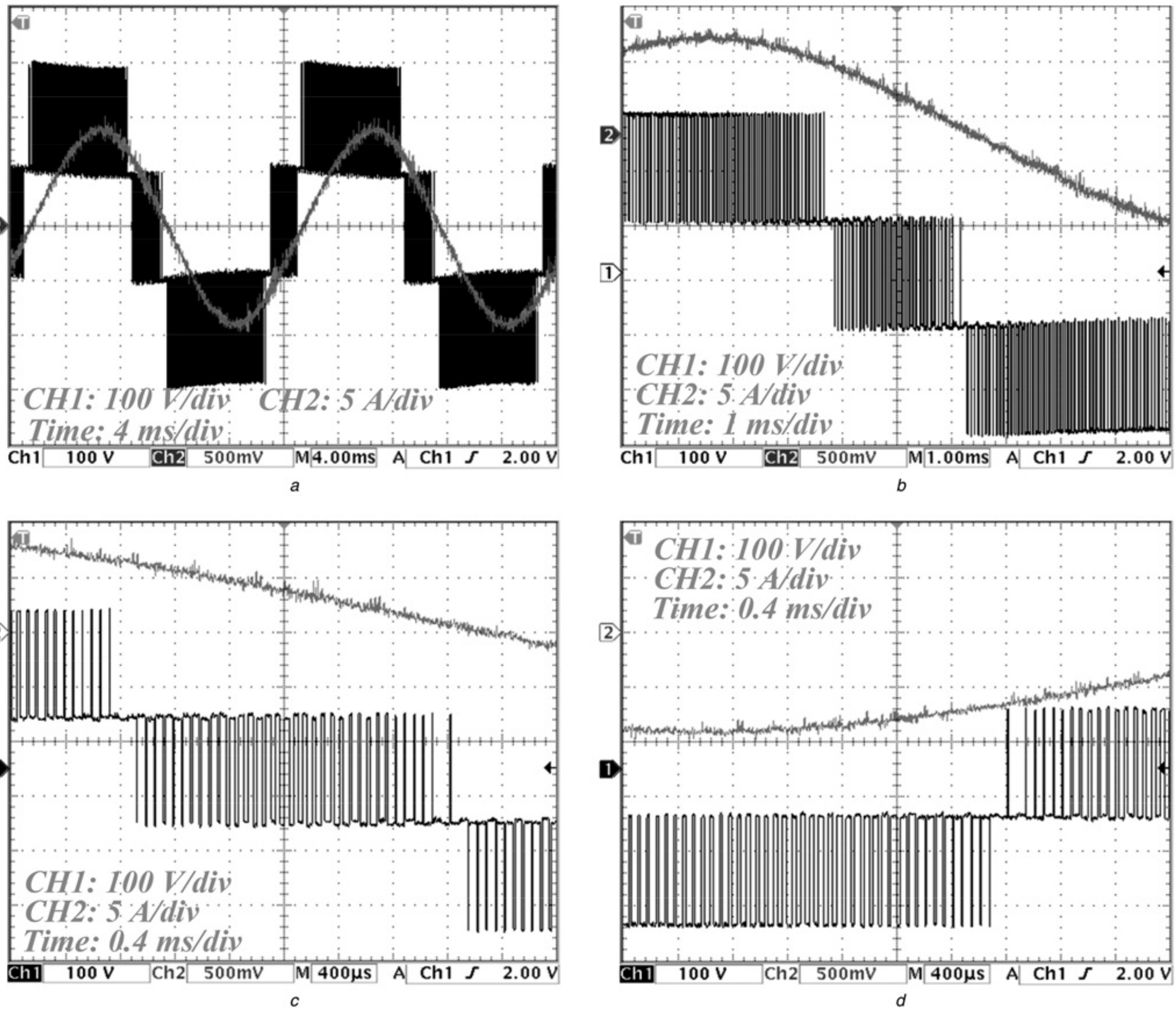


Fig. 5 *Experimental results of the proposed 3-cell-four-level FCMC:*

a Two cycles of output voltage and load current
b Half-cycle of output voltage and load current
c Closer view of output voltage and load current
d Detailed observation on output voltage and load current

doi: 10.1049/iet-pel.2013.0071

where f_{SW} is the switching frequency, L_b and C_b are the inductance and the capacitance of the output RLC balance booster circuit, respectively.

3 Proposed flying capacitor multicell converter

As shown in Fig. 2, the conventional configuration of a 4-cell-five-level FCMC requires two dc voltage sources whereas the upper dc voltage source is utilised when the power switch S_4 conducts, that is, it is closed and its state is 1, and the lower dc voltage source is utilised when the power switch $\overline{S_4}$ conducts. As a result, each of two dc voltage sources is only required as its associated power switch conducts. It means that two dc voltage sources are not conducting simultaneously. This fact makes it possible to modify the circuit topology by eliminating one of the dc voltage sources.

To settle the aforesaid logic and to aim the elimination of one of the dc voltage sources, two auxiliary power switches with complementary states to each other (A and $\overline{A}$) must be added to the configuration of the FCMC in order to provide an appropriate path for ac current afterward removing the converter's midpoint. Such amendment to the FCMC topology is attained on the basis that FCs have got stabilised at their nominal values. Hence, the modified 4-cell-five-level circuit of the FCMC topology can be achieved in a structure which is represented in Fig. 3*a* as a first design.

After equipping the FCMC with two power switches (A and, $\overline{A}$) and removing one of the dc voltage sources the direction of power of switches of S_4 and $\overline{S_4}$ must be reversed because of the fact that the voltage rating of capacitor C_4 in cell-4 (i.e. $E_{C4} = 3E/4$) is bigger than the voltage rating of main dc-link ($E/2$) in the proposed configuration in steady stare.

However, it is worth mentioning that E_{C4} is smaller than the main dc-link voltage ($E/2$) during the start-up transient. It means that after connecting the FCMC to the dc voltage bus and hypothesising that the FCs have not been pre-charged to their nominal dc voltages, natural balancing will start to charge and regulate the E_{C4} at its steady state value ($3E/4$) inherently. Meanwhile, E_{C4} would be smaller than the main dc–link ($E/2$) at the start-up, that is, voltage build up period, and subsequent to proceeding transient state while being slowly charged by the positive current flowing through the FC of C_4, it would reach to $E/2$ and afterward it would pass $E/2$ to reach to $3E/4$ to obtain stabilised at its nominal dc voltage. Ultimately, E_{C4} would be bigger than the main dc-link voltage ($E/2$).

The aforesaid phenomenon implies that during the voltage build up transient it is mandatory to retain the direction of the power switches of S_4 and $\overline{S_4}$ (in general, the direction of the power switches situated in the immediate vicinity of the dc voltage source) in their normal position whereas they must be reversed and inverted after proceeding the transient state.

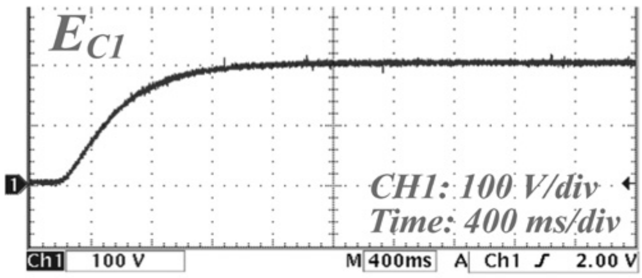
As a result, the bidirectional power switches must be used for power switches of S_4 and $\overline{S_4}$ in the proposed FCMC topology. Hence, the second design of the proposed FCMC can be achieved in a structure which is represented in Fig. 3*b* wherein another pair of complementary power switches (B and $\overline{B}$) with opposite directions to S_4 and $\overline{S_4}$ are embedded to the circuit. However, the switches B and $\overline{B}$ can always stay ON after proceeding the start-up transient time.

Moreover, Fig. 3*c* illustrates the general configuration of n-cell-n + 1-level proposed FCMC. Moreover, as an example, Fig. 4 presents the switching strategy, states of power switches and the output voltage of the 4-cell-five-level proposed FCMC which is controlled and operated by an amended PSC-PWM technique and a modulation index equal to 0.8 (M = 0.8).

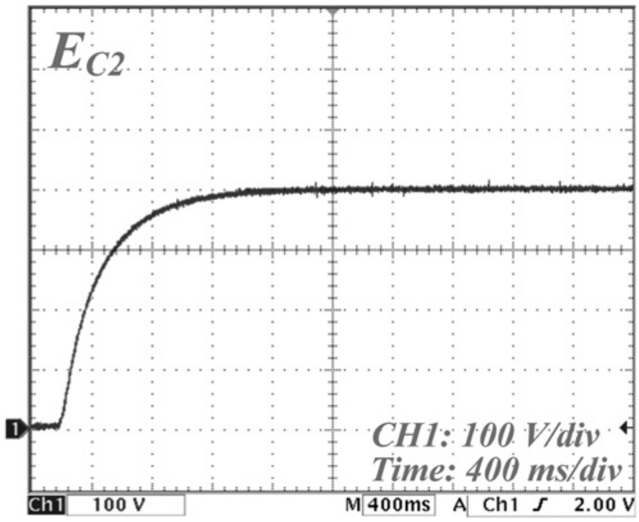
In Fig. 4, the power switch is ON when its state is 1 and is OFF when its state is 0. Also, states of power switches in the 4-cell-five-level proposed FCMC have been demonstrated in Table 2 wherein the voltage variation of each FC, that is, $\Delta E_{Cx} = E_{Cx}(t + \Delta t) - E_{Cx}(t)$, is determined for each switching state by taking into account the direction of the load current.

Comparing Figs. 1 and 3*c* reveals that the conventional FCM requires dc voltage source with total voltage rating of E to produce an output voltage with peak to peak value of E whereas the proposed VSMMC just needs one dc voltage source with halved voltage rating ($E/2$) to produce the same output voltage (output switched voltage with peak to peak value of E).

Elimination of one dc voltage source optimises the structure of FCMC from size and cost points of view which will be discussed later. Other parameters of the proposed FCMC such as the voltage rating of FCs and power switches are as the same as those in the conventional topology.



a



b

Fig. 6 *Experimental observations about the FC voltages pertinent to the proposed 3-cell-four-level FCMC:*

a E_{C1}
b E_{C2}

doi: 10.1049/iet-pel.2013.0071

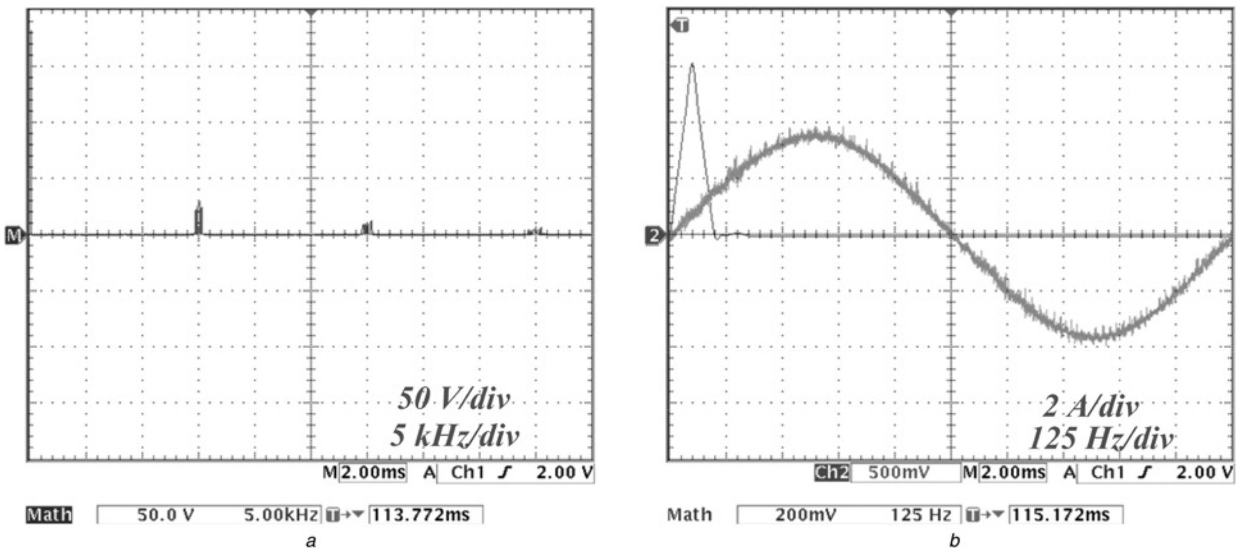


Fig. 7 *Experimental results of the proposed 3-cell-four-level FCMC: frequency spectrum*

a Output voltage
b Load current

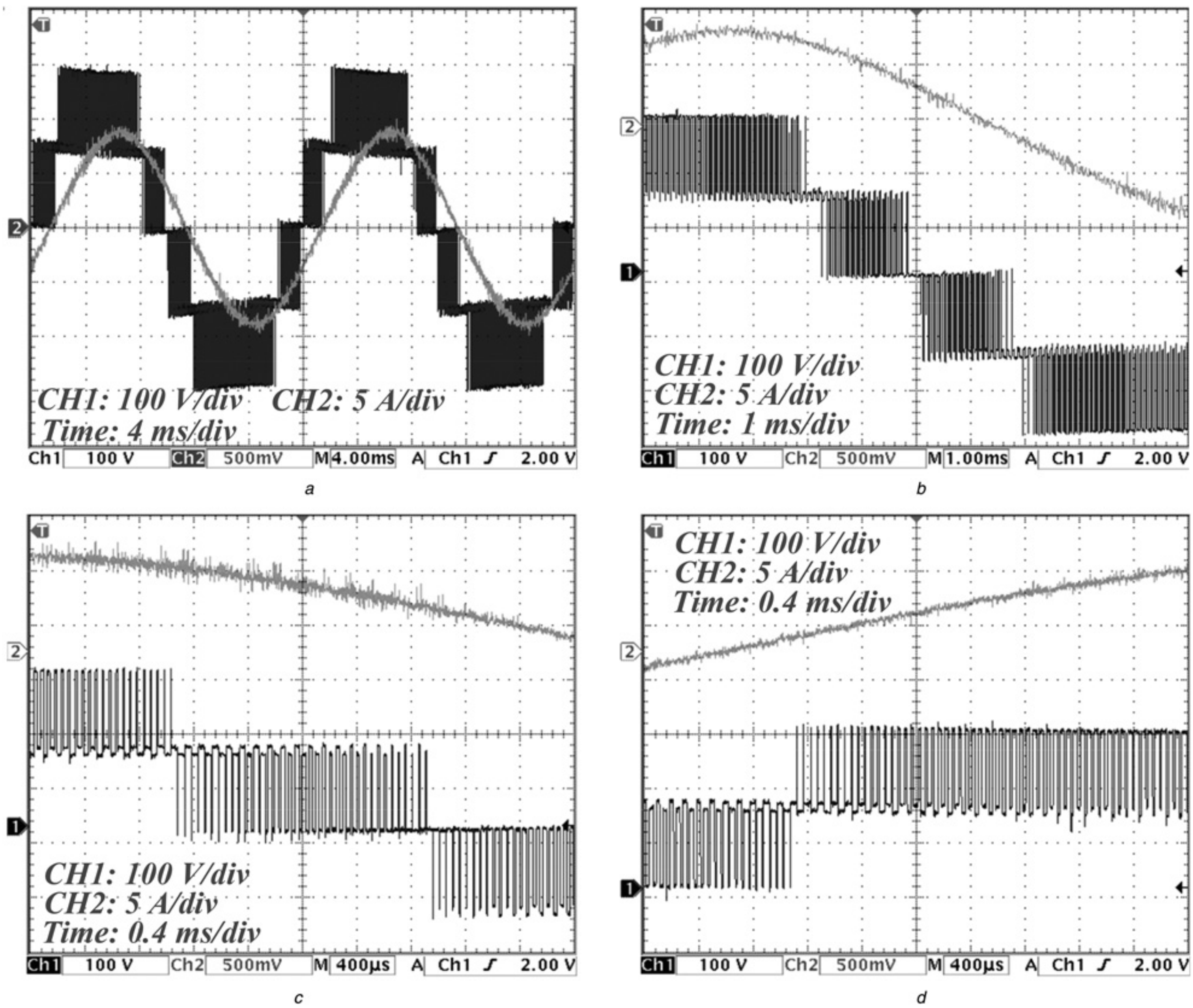


Fig. 8 *Experimental results of the proposed 4-cell-five-level FCMC:*

a Two cycles of output voltage and load current
b Half-cycle of output voltage and load current
c Closer view of output voltage and load current
d Detailed observation on output voltage and load current

4 Experimental observations of the proposed FCMC

To attest both the validity and viability of the proposed topology, 3-cell-four-level and 4-cell-five-level enhanced FCMC prototypes are built utilising Siemens BUP 314D 1200 V 42 A IGBTs. The measured output voltage and load current, FC voltages and frequency spectrum of output voltage and load current taken from the 3-cell-four-level prototype with the parameters abridged in Table 3 are presented in Figs. 5–7.

As shown in Fig. 6, the voltages of FC are stabilised and fixed at their requisite values, that is, 200 and 400 V, without any feedback signal in spite of elimination of one of the dc voltage sources. This issue demonstrates the natural balancing property in the proposed FCMC.

Another experimental study is accomplished on the 4-cell-five-level proposed FCMC with the same parameters listed in Table 3. The experimentally measured output voltage, load current, FC voltages and frequency spectrum of the output voltage and load current taken from the 4-cell-five-level prototype are presented in Figs. 8–10. As shown in Fig. 9, the voltages of FCs are stabilised and fixed at their requisite values, that is, 150, 300 and 450 V, without any feedback signal which corroborate the natural balancing property in the proposed FCMC.

It is worth mentioning that the voltage rating of dc voltage source used in both experiment of 3-cell and 4-cell proposed FCMCs is 300 V whereas two 300 V dc voltage sources (total voltage rating of 600 V) are required in conventional FCMC to produce the same output voltages with the same FC voltages.

It is worth mentioning that output voltage of proposed n-cell FCMC as such as the conventional topology has group harmonics around the $(n \times k \times f_{SW})$th harmonic where k and f_{SW} are an integer number and the switching frequency, respectively [15, 16]. As shown in Figs. 7*a* and 10*a*, output voltage of the 3-cell and 4-cell proposed FCMCs has harmonic groups around integer multiples of 15 and 20 kHz, respectively, whereas switching frequency is 5 kHz.

Furthermore, experimental studies are accomplished on an 8-cell-nine-level prototype system of the proposed converter with the abridged parameters in Table 3 in order to investigate its feasibility for the higher number of cells and voltage levels. The results are presented in Fig. 11.

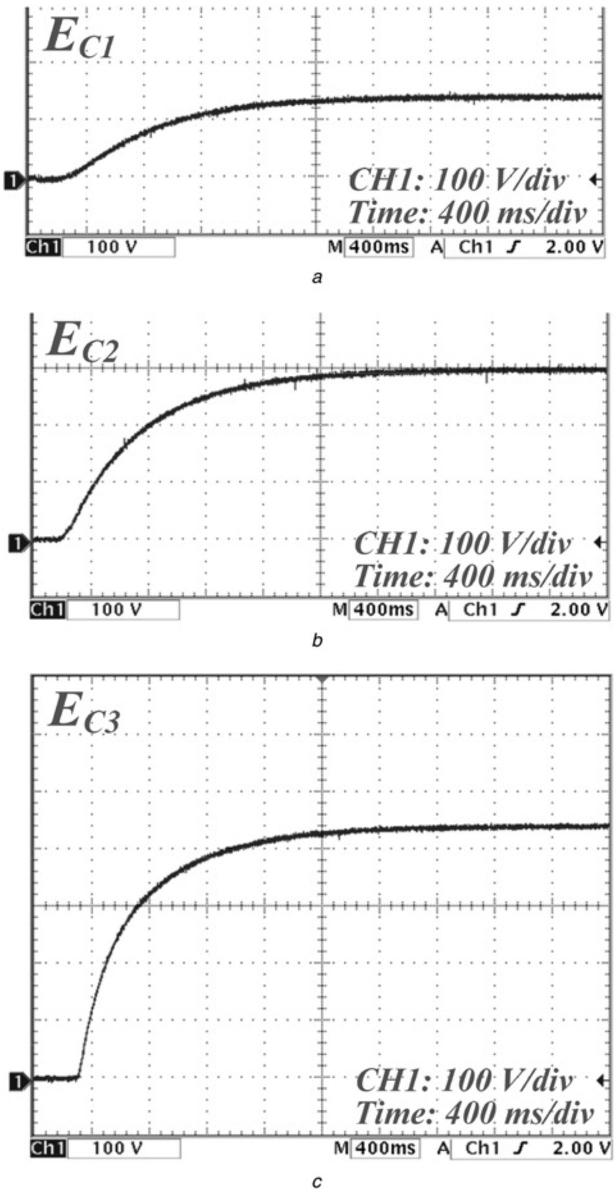


Fig. 9 *Experimental observations about the FC voltages pertinent to the proposed 4-cell-five-level FCMC:*

a E_{C1}
b E_{C2}
c E_{C3}

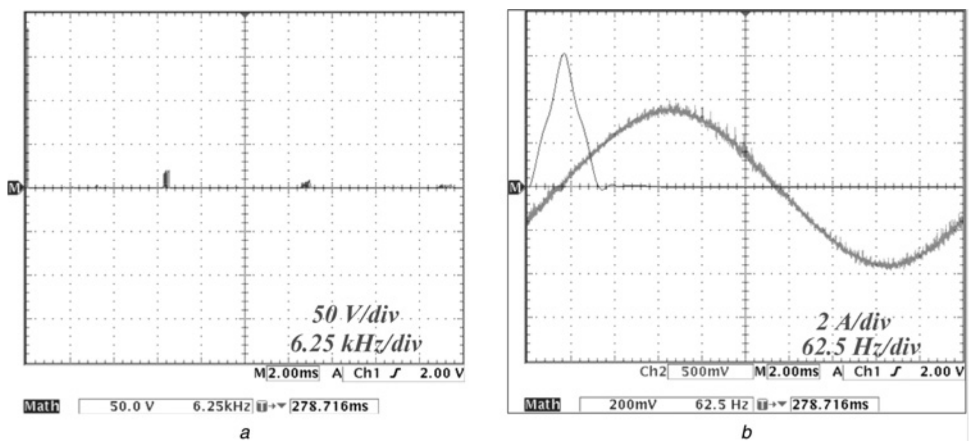


Fig. 10 *Experimental results of the proposed 4-cell-five-level FCMC: frequency spectrum*

a Output voltage
b Load current

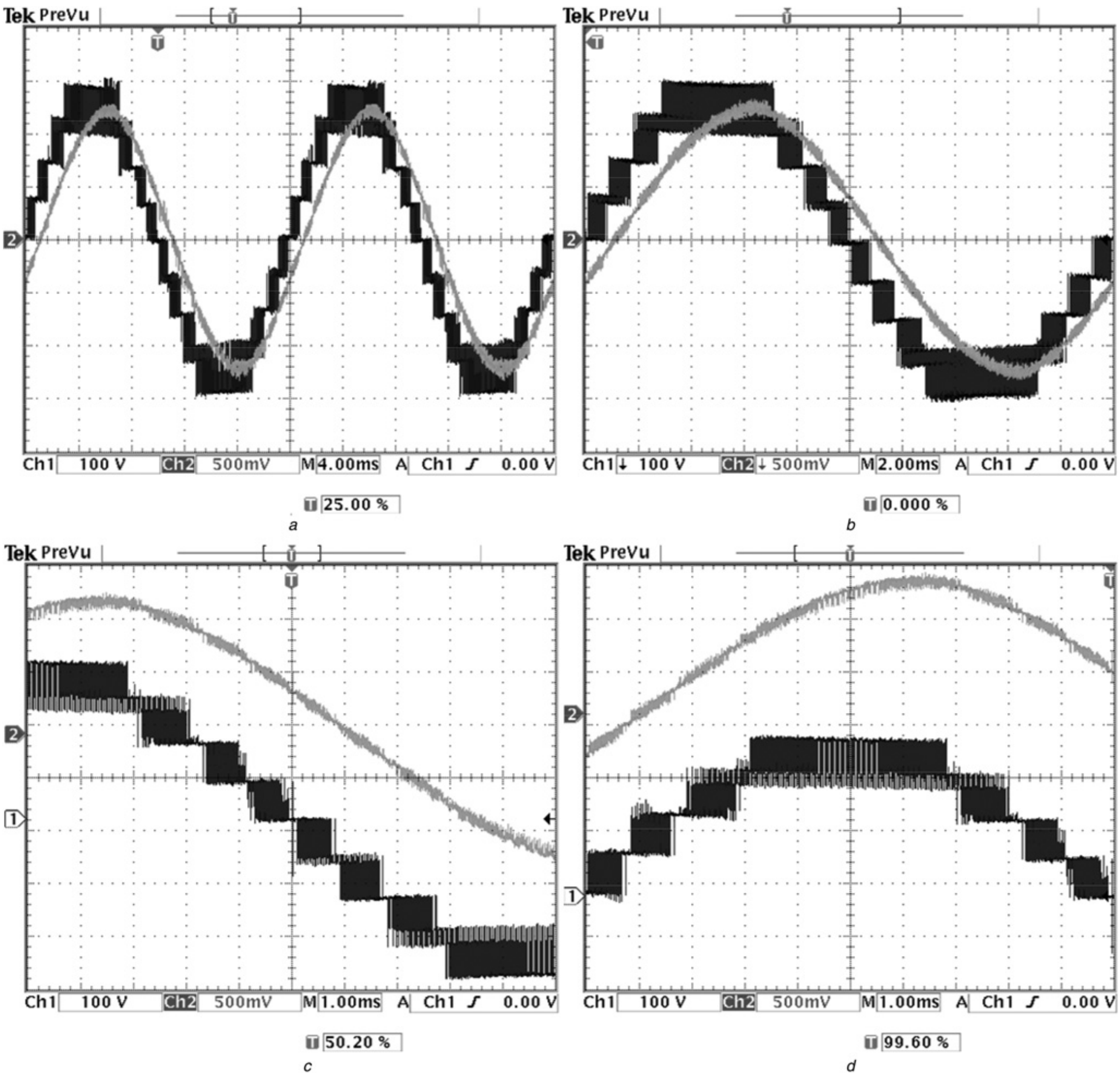


Fig. 11 *Experimental results of the proposed 8-cell-nine-level FCMC:*

a Two cycles of output voltage and load current
b One cycle of output voltage and load current
c Half-cycle of output voltage and load current
d Detailed observation on output voltage and load current

5 Price comparison of the conventional and proposed FCMCs

As discussed in previous sections, the main advantage of proposed FCMC in comparison with conventional one is that the required number of capacitors in main dc-link is one in proposed converter instead of two in conventional topology. In other words, the number of capacitors in main dc link of proposed FCMC is decreased from 2 to 1 whereas four switches are added; however, the number of other components such as FCs and cells' switches as well as their specification such as voltage rating and current rating are same in both proposed and conventional FCMCs for same power rating and condition. It means that the only difference between the proposed converter and conventional one is that the proposed one has one less capacitor in main dc link and four more switches.

Here, the comparison is done between two converters from price point of view. As an example, 15 kW conventional and proposed converters based on configurations shown in Fig. 12 are simulated to obtain the current rating of four extra switches (a_1, a_2, b_1, b_2) and dc-link capacitors. In Fig. 12, the output voltage of converters is ± 450 V peak, output current is 45 A rms, resistive-inductive load is 5.5 Ω–13 mH, resistive load is 6.8 Ω, switching frequency is 5 kHz, modulation index is 0.95 and input filter inductance is 2 mH. The reason for selecting the voltage of 450 V for dc-link capacitors is that 450 V is the maximum voltage of dc capacitors whose price are available in online vendors such as NEWARK, DIGIKEY and MOUSER or the website of some companies such as IRF and ABB. According to simulation results, the rms current of each of two dc capacitors (upper and lower) in Fig. 12*a* is almost 31 A whereas in Fig. 12*b*, the rms current of one dc-link capacitor is 34 A, rms current of switches a_1 and a_2 is 30 A and current rating of switches b_1 and b_2 is ~ 10 A for both inductive-resistive and purely resistive loads.

Based on the obtained current rating of dc capacitors in both converters, some 450 V capacitors from different manufacturers are selected and illustrated with their price in Table 4 to satisfy the required specification (online vendors). For example, to have 31 A in each of upper and lower side dc-link capacitors in conventional one, four capacitors of CGH312T450W5 L (Cornell Dubilier) with current rating of 8.5 A can be connected in parallel in each of upper and lower side to be able to handle 31 A (rms); thus, total of 8 (2×4) capacitors are required in conventional FCMC; whereas only four of the same capacitor should be connected in parallel to handle the current of 34 A (rms) in proposed FCMC.

On other side, the price of 600 V IGBT switches with current rating of 30 A (even higher) at case temperature of 85° such as IRG4PC40FPBF, IRG4PC40SPBF, IRG4PC50UPBF (all form INTERNATIONAL RECTIFIER), IGW50N60T (from INFINEON), HGTG30N60B3 and HGTP12N60A4D (from FAIRCHILD SEMICONDUCTOR) to be used as four extra switches in proposed FCMC, is almost \$3–5 (online vendors). By considering the driver IC, heat sink and isolated power supply for each switch, the total price for each switch is around \$15–20.

Thus, the total price of four additional switches in proposed converter is almost \$60–80. It should be mentioned that both current and voltage rating of switches b_1 and b_2 is less than the current and voltage rating of switches a_1 and a_2 while the same switch is selected for all four extra switches. Moreover, there is no need to have separated driver IC plus its own isolated power supply for each of switches a_1 and a_2 where boot strap capacitor can be used. Therefore considering these two facts can reduce the estimated \$60–80 price for additional switches. Moreover, in Figs. 12*a* and 12*b*, the total price of isolated transformers (because of same power rating in both configurations), diode rectifiers and input filter inductances in both conventional and proposed converters is almost same.

Finally, it can be concluded that the price of proposed converter is \$160 (in worst case) to \$340 (in best case) cheaper than conventional FCMC. It is worth mentioning that selected capacitors exactly have the same voltage rating of dc link and same current rating obtained from simulation while there should be some margins between the required specifications and specification of selected components

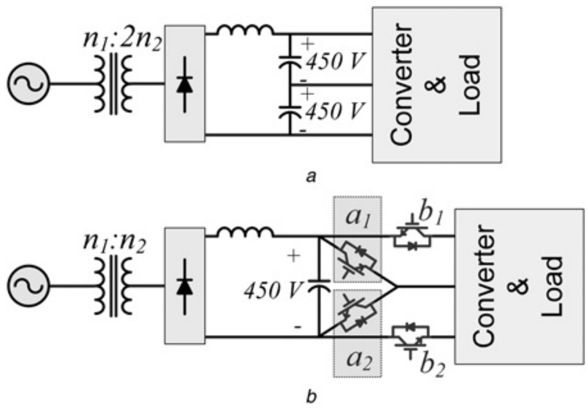


Fig. 12 *Configuration of simulated circuits to obtain current rating of components for price comparison*
a Conventional FCMC
b Proposed FCMC

(voltage rating margin is considered for additional switches); this issue results increasing the price difference.

Moreover, the life time of dc capacitors is 8000–12 000 h (0.9–1.4 year) which causes replacing dc capacitors after a while; as a result, proposed converter can save much more money in long term application which is essential in industrial applications.

Furthermore, the size and weight of mentioned capacitors are significantly higher than four additional switches, including their heat sink, driver etc., which cause reduction of size and weight of proposed FCMC in comparison with conventional FCMC with same power rating and specifications. Here, the price of conventional and proposed FCMCs with specification of 15 kW 320 V rms is compared just as an example to show the potential of proposed converter to have cheaper price.

Definitely, the price, size and weight difference of compared converters would be much more significant in high-power medium-voltage application in which the price and size of dc capacitors increases rapidly.

6 Comparison of other types of multilevel converters with proposed FCMC

Although, the major part of this paper is devoted to analyse the conventional and the proposed FCMCs, the comparison between the different types of multilevel converters such as NPC, CM, FCM and proposed FCM for producing the identical output voltage with equal number of levels (i.e. $2n + 1$ levels and $2E$ peak to peak) is illustrated in Table 5

Table 4 Price comparison of the conventional and proposed FCMC from main dc-link capacitors point of view

Model (manufacturer)	RMS ripple current (A) at 85° and 50 Hz	Capacitance, F	Price, \$	Required number in converter		Price difference (\$) (conventional − new)
				Conventional (upper + lower)	Proposed	
LNC2W332MSEG (NICHICON)	12	3300	85	6	3	+ 255
LNY2W472MSEG (NICHICON)	12	4700	100	6	3	+ 300
DCMC472T450DE2B (Cornell Dubilier)	10	4700	110	6	4	+ 220
CGH312T450W5 L (Cornell Dubilier)	8.5	3100	100	8	4	+ 400

doi: 10.1049/iet-pel.2013.0071

Table 5 Comparison between the different types of multilevel converters regarding number of components in producing the identical output voltage (i.e. $2n+1$ levels, $2E$ peak to peak)

Type of multilevel converter	No. of high frequency switches	No. of isolated DC sources	Voltage rating of DC sources, per unit	No. of flying capacitors	No. of isolation transformer and diode bridge rectifier	No. of clamping diode with the identical voltage ratings
NPC	$4n$	1	$2n$	0	1	$4n^2-2n$
CM	$4n$	n	1	0	n	0
FC	$4n$	1	$2n$	$2n-1$	1	0
proposed FC	$4n+4$	1	n	$2n-1$	1	0

Table 6 Advantages and disadvantages of different types of multilevel converters

Type of multilevel converter	Advantage	Disadvantage
NPC	1) Requires just one isolated dc link 2) Voltage of capacitors in main dc link is identical	1) Requires voltage sensor for each capacitor and one current sensor if number of levels is more than 3 2) Does not have self-balancing property 3) Rapid growth in number of clamping diodes by increasing the number of levels 4) Complex control method to balance the voltage of dc-link capacitors
CM	1) Does not require any voltage or current sensor 2) Modular configuration	1) Requires bulky and costly isolated transformer for each cell (in power conversions involving active power) 2) Requires voltage sensor for each cell (in power conversions involving just reactive power)
Conventional and proposed FCM	1) Does not require any voltage and current sensor 2) Self-balancing property of the FC without any feedback control	Voltage of the FC in each cell is different. Thus, needs bulky and costly dc capacitors

whereas advantages and disadvantages of each configuration is presented in Table 6.

7 Conclusion

This paper has proposed a novel topology for FCMCs. The main advantage of the proposed converter, in comparison with conventional configuration of FCMC, is a reduction in the number of required dc-voltage sources from two to one whereas the voltage rating of FCs and high-frequency switches are kept unaltered. This progress is achieved by adding four power switches (IGBTs) to the conventional configuration of an FCMC. Incontrovertibly, diminution one of the required dc voltage sources in the FCMCs abates their size, weight and cost which is analysed and proved in Section 5. In addition, the natural balancing property of the FCMCs is preserved in the proposed topology since its control strategy is based on the PSC-PWM. Provided experimental results pertaining to the enhanced 4, 5 and 8-level FCMCs corroborate the outstanding performance, viability and feasibility of the proposed FCM converter, its suggested switching and modulation methodologies specifically for high power applications.

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doi: 10.1049/iet-pel.2013.0071